Missing & Censored Data

Group 4

STAT 5440 • Spring 2026

Outline

1. Motivation

2. Missing Data

3. Censor Data

4. Summary

Motivation

Why model the data collection mechanism?

- We rarely want conclusions about the data in our spreadsheet, we want conclusions about the population behind it
- However, dataset might have:
  - **Unrecorded** values
  - **Non-sampled** units

Then, the process that generated the observed sample is *part of the model*.

Example

Survey: "What is your income?"

You receive $n = 1{,}000$ responses.

But 300 people *refused to answer*.

If you analyze only the 700 complete rows:

- Your sample mean is a valid estimator *of who responded*.
- Not of the population, unless non-response is random.
- However, very high or very low earners are the most likely to decline, biasing our results.

General Data Framework

Observed vs. Complete Data

Setting	**Observed**	**Complete (unobservable ideal)**
Sampling	n units in the sample	N units in the population
Experiment	Outcomes of *treated* units	Outcomes under *all* treatments
Rounded data	Rounded observations	Precise values
Missing data	Recorded values	All values (observed + missing)

Bayesian view: treat whatever is unobserved as a latent quantity with a prior, and let the posterior integrate it out.

Two Motivating Problems

Problem 1: Missing data.

- NHANES adult survey. Outcome: systolic blood pressure.
- Predictor BMI missing for 8% of respondents.
- Older respondents skip BMI more often than younger ones.

Problem 2: Censored data.

- Lung cancer study. Outcome: time to death.
- 28% of patients alive at study end.
- For those rows, survival time is only known to exceed follow-up.

Common structure: every unit generated a complete Y_i. We observe only part of it.

A Unified Bayesian View

Recipe (both problems):

1. **Augment** the parameter space with the unseen quantities.
2. **Constrain** them with the partial information we have.
3. **Propagate** uncertainty to the parameters of interest.

What the partial information is:

- *Missing data*: whatever correlated variables tell us.
- *Censored data*: a set containing the value (e.g., $Y_i > c_i$).

Censored data is an informative special case of missing data.

Missing Data

The missing data problem, formally

Ingredients

- Complete data $y = (y_{obs}, y_{mis})$
- Inclusion indicator I with $I_{ij} = 1$ if observed, 0 if missing
- Data parameter θ
- Missingness parameter ϕ

Joint model factors into two pieces:

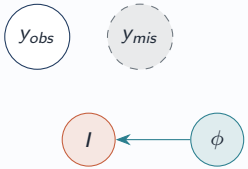
$$\underbrace{p(y, I \mid \theta, \phi)}_{\text{what we model}} = \underbrace{p(y \mid \theta)}_{\text{data-generating process}} \cdot \underbrace{p(I \mid y, \phi)}_{\text{missingness mechanism}}$$

But we only see (y_{obs}, I) — so the observed-data distribution is

$$p(y_{obs}, I \mid \theta, \phi) = \int p(y_{obs}, y_{mis} \mid \theta)\, p(I \mid y_{obs}, y_{mis}, \phi)\, dy_{mis}$$

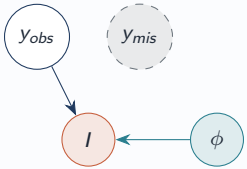
Types of Missing Data

MCAR



$p(I \mid y, \phi) = p(I \mid \phi)$

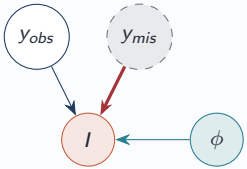
MAR



I depends on *observed* data only.

$p(I \mid y, \phi) = p(I \mid y_{obs}, \phi)$

MNAR



Missingness depends on the missing value itself.

$p(I \mid y, \phi) = p(I \mid y_{obs}, y_{mis}, \phi)$

Ignorable: If is MAR and θ is independent of ϕ.
Under ignorability, we can forget about $p(I \mid \cdot)$ and just model $p(y_{obs} \mid \theta)$.

Example

Setup. We collect $y = (\text{age}, \text{ income})$. Age is always recorded; income is sometimes missing.

Regime	Why income is missing	What it looks like
MCAR	A random 10% of surveys get lost in the mail.	Missing rate flat across all ages and incomes.
MAR	Older respondents skip the money question more often.	Missing rate rises with *age* (observed).
MNAR	High earners refuse to disclose income.	Missing rate depends on *income itself* (unobserved).

MCAR and MAR $\Rightarrow$ standard Bayesian model on y_{obs} is fine.
MNAR $\Rightarrow$ you *must* model $p(I \mid y_{mis})$.

Case Analysis

Default reflex: drop every row with any `NA`, fit as usual.

Under MCAR: unbiased. Wastes partially complete rows.

Under MAR or MNAR: biased. The complete-case sample is selected, not random.

Goal: use every row, and propagate the extra uncertainty for imputed values.

Solutions when the missing mechanism is not ignorable

1. Multiple Imputation (MI)

1. Build imputation model $p(y_{mis} \mid y_{obs})$
2. Draw K completed datasets $y^{(1)}, \ldots, y^{(K)}$
3. Fit the analysis model *separately* to each
4. Pool the K posteriors (Rubin's rules)

Great when: many variables with mixed types, easier with R-packages like `mice`

2. Data augmentation (in Stan)

- Declare `y_mis` in `parameters`
- Reassemble full `y` in `transformed parameters`
- In `model`, write *both* the imputation model *and* the analysis model

Great when: already in Stan, want one coherent joint posterior, care about propagating all uncertainty automatically.

Chained-Equation Imputation (`mice`)

Fully Conditional Specification:

1. For each incomplete Y_j, specify a conditional $p(Y_j \mid Y_{-j})$ (typically a small regression).
2. Initialize missing cells with simple fills.
3. Cycle through variables: draw $Y_j \mid Y_{-j}$ for every incomplete j.
4. Iterate to convergence. Save K completed datasets spaced along the chain.

Strengths: handles mixed variable types; each conditional is small.

Caveat: the joint implied by the conditionals may not be self-consistent (the *compatibility problem*).

Congeniality

Principle (Meng, 1994): the imputation model must be at least as rich as the analysis model.

Failure mode:

- Analysis model contains interaction $X_1 \times X_2$.
- Imputation model for $X_1 \mid X_2$ is purely linear.
- Imputations smooth out the interaction. The analysis cannot recover it.

Rule: include every variable and every nonlinearity the analysis model uses. When in doubt, include more.

Example: Setup

Goal: Estimate the relationship between age and income when income is sometimes missing.

- **Data:** Simulated sample of $n = 500$ with income $\approx 20 + 0.8 \cdot \text{age} + \text{noise}$
- **Missingness mechanism:** Probability of missing income increases with age — older respondents are more likely to skip the question
- This is **MAR**: missingness depends on age (observed), not on income itself
- **Plan:**
  - Use `mice` to generate $K = 5$ imputed datasets
  - Fit `lm(income ~ age)` on each
  - Combine with `pool()` (Rubin's rules)

Example: R code

```
library(mice)
set.seed(123)

# Simulate age/income with MAR missingness
n <- 500
age <- round(runif(n, 25, 65))
income <- 20 + 0.8 * age + rnorm(n, 0, 10)
income[runif(n) < plogis(-3 + 0.05*age)] <- NA
dat <- data.frame(age, income)

# Multiple imputation (K = 5 datasets)
imp <- mice(dat, m = 5, printFlag = FALSE)

# Fit model on each, combine via Rubin's rules
fit    <- with(imp, lm(income ~ age))
pooled <- pool(fit)
summary(pooled)
```

Visualizing Multiple Imputation Results

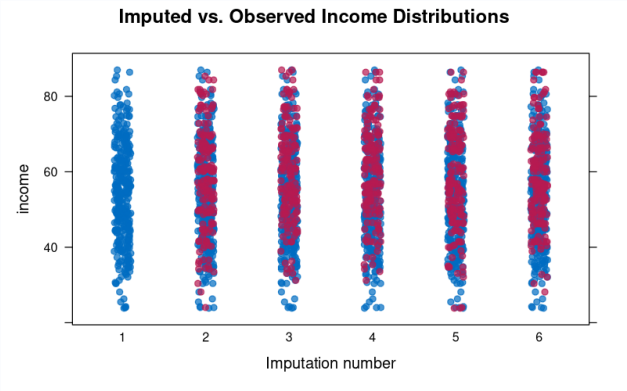


Figure 1: Imputed income (red) vs. observed (blue). Distributional consistency suggests a valid imputation model.

Figure 2: Pooled fit. The light blue lines show the $K = 5$ individual regressions; the dashed line is the final pooled estimate.

MNAR Models: Selection and Pattern-Mixture

Under MNAR, the mechanism depends on y_{mis} and cannot be ignored.

Selection models: $p(y, I) = p(y \mid \theta)\, p(I \mid y, \phi)$. Model $P(\text{missing})$ as a function of y_{mis}.

Pattern-mixture models: $p(y, I) = p(y \mid I, \theta_I)\, p(I \mid \phi)$. Separate outcome distribution per missingness pattern. Natural for dropout: completers and dropouts are two populations.

Reporting: MNAR is untestable. Sensitivity analysis across MNAR assumptions replaces goodness of fit.

Applied Example: NHANES

Data: US National Health and Nutrition Examination Survey 2009–2012, via the `NHANES` R package. Roughly 7,150 adult respondents.

Question: How does total cholesterol associate with systolic blood pressure, adjusting for age and BMI?

Variables:

- `BPSysAve`: average systolic BP (outcome).
- `Age`, `BMI`: fully observed.
- `TotChol`: total cholesterol (mmol/L), missing for 5% of rows.
- `Weight`, `Gender`: auxiliary predictors for the imputation model.

Missingness assumption: MAR given age, BMI, weight, and gender. Revisited in the sensitivity slide.

Priors and Prior Predictive

Outcome model (SBP in mmHg):

$$\text{BPSysAve}_i = \alpha + \beta_{\text{age}}\text{Age}_i + \beta_{\text{BMI}}\text{BMI}_i + \beta_{\text{TC}}\text{TC}_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

$$\alpha \sim \mathcal{N}(120, 20^2), \quad \beta_{\text{age}} \sim \mathcal{N}(0, 1^2),$$
$$\beta_{\text{BMI}} \sim \mathcal{N}(0, 5^2), \quad \beta_{\text{TC}} \sim \mathcal{N}(0, 5^2), \quad \sigma \sim \mathcal{N}^+(0, 10^2).$$

Imputation model (TotChol in mmol/L):

$$\text{TC}_i = \gamma_0 + \gamma_{\text{age}}\text{Age}_i + \gamma_{\text{BMI}}\text{BMI}_i + \gamma_{\text{wt}}\text{Wt}_i + \gamma_{\text{gen}}\text{Gen}_i + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma^2_{\text{TC}})$$

$$\gamma_0 \sim \mathcal{N}(5, 2^2), \quad \gamma_\cdot \sim \mathcal{N}(0, 1^2), \quad \sigma_{\text{TC}} \sim \mathcal{N}^+(0, 2^2).$$

Fit and MCMC Diagnostics

Stan data-augmentation model. 4 chains, 2000 iterations, warmup 1000.

Convergence on outcome-model parameters:

- $\widehat{R} \leq 1.005$ for α, β_{age}, β_{BMI}, β_{TC}, σ.
- Effective sample size $>$ 2,700 out of 4,000 draws for all parameters of interest.
- Trace plots show good mixing across all four chains.

Posterior Summaries

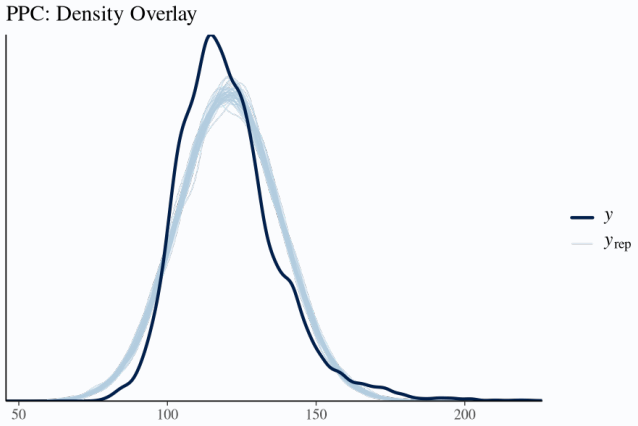
Posterior means and 95% credible intervals:

Parameter	Mean	95% CrI	Units
α	107.32	(104.96, 109.66)	mmHg
β_{age}	0.39	(0.37, 0.41)	mmHg / year
β_{BMI}	0.28	(0.22, 0.33)	mmHg / BMI unit
β_{TC}	1.08	(0.73, 1.43)	mmHg / (mmol/L)
σ	15.32	(15.08, 15.56)	mmHg

At fixed age and BMI, each 1 mmol/L increase in total cholesterol associates with a 1.08 mmHg increase in mean SBP. The posterior for β_{TC} excludes zero by 6.2 posterior standard deviations.

Data augmentation also yields posterior distributions for the 356 imputed TotChol values, which `mice` discards after pooling.

Posterior Predictive Check



The model recovers the bulk of the distribution; the upper tail is slightly underdispersed.

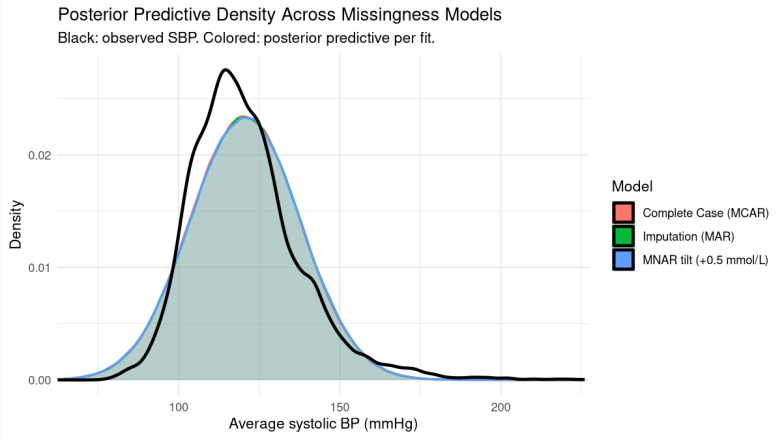
Sensitivity to the Missingness Assumption

Three fits:

Missingness model	Posterior mean β_{TC}	95% CrI
Complete case (implicit MCAR)	1.08	(0.73, 1.44)
Imputation (MAR)	1.08	(0.73, 1.43)
MNAR tilt: $\text{TC}_{\text{mis}} + 0.5$ mmol/L	1.18	(0.83, 1.54)

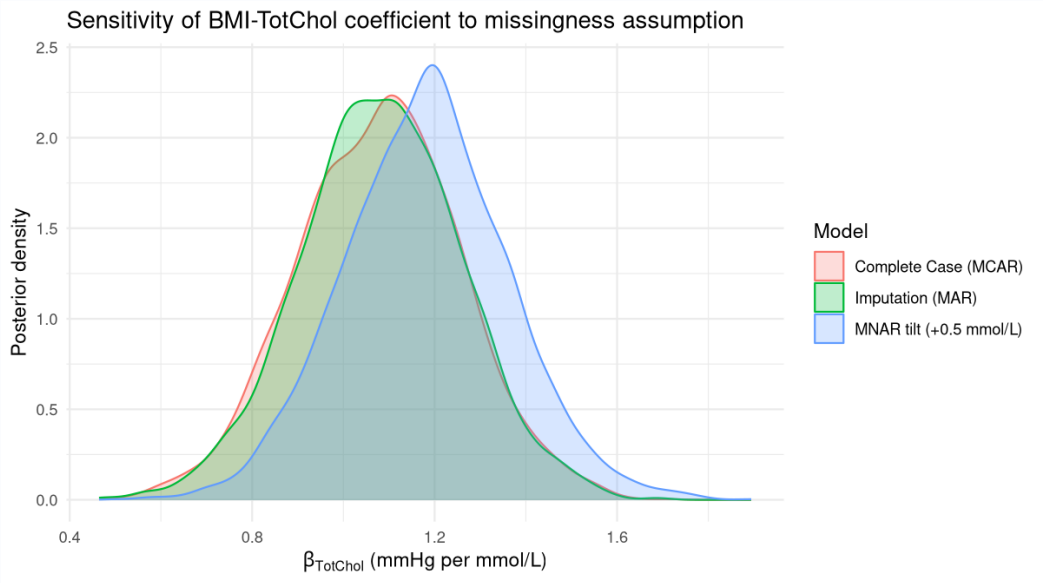
The positive cholesterol–SBP association holds under all three assumptions. The MNAR tilt shifts the point estimate by roughly 0.10 mmHg per mmol/L, about half a posterior SD. Conclusion is robust to MAR departures of this magnitude.

Predictive Fit Across Missingness Models



Marginal predictive density is indistinguishable across MCAR, MAR, and MNAR fits.

Beta Comparison for Different Models



Censor Data

Censor data

Definition

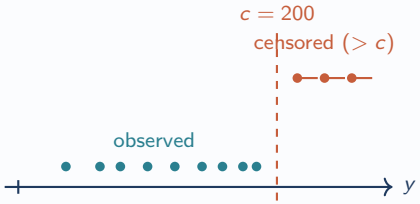
Observation outcomes are known to fall in some interval (a, b), but they are unknown where in that interval they lie.

- Right censoring (a, $+\infty$)
  - Example: 12-month study tracking how long a specific laptop battery lasts
- Left censoring ($-\infty$, b)
  - Example: Water quality test for a specific pollutant
- Interval censoring (a, b)
  - Example: Dentist checking patients for cavities

Censor data- Example

We are interested in the average weight of a population, and we take a sample of 100 individuals.

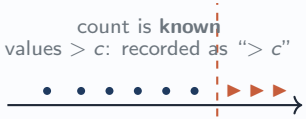
- The scale tops out at 200 kg.
- 91 measurements come back as numbers.
- 9 come back as "over 200".



We don't know what those 9 weights are — but we know there were 9 of them, and we know each is > 200.

Censoring vs. Truncation

Censoring



We know: # censored, threshold.
We don't know: exact value.

Truncation

We know: threshold.
We don't know: # truncated, nor exact values.

In truncated data, observations beyond the threshold are lost completely, and their total count is unobserved

Motivation

What problem does this method solve?

It enables valid inference when outcomes are censored by either modeling them as unknown but constrained values or integrate over possible values in the likelihood function instead of treating them as exact data.

Why can't standard regression handle censored data?

Standard regression assumes exact y_i, so treating censored values as fixed numbers both biases estimates and uses the wrong likelihood, ignoring that the true value is only known within a range.

Where does it show up in practice?

Censored data appears whenever values are only partially observed, such as survival analysis which is conducted for a limited time period, measurements with detection limits.

Potential Solutions

1. Marginalization:

- No need to explicitly sample missing values
- Integrate out censored values in the likelihood
- Use CDF terms

2.Data augmentation (latent variables)

- Treat censored y_i as unknown parameters
- Constrain them (e.g., $y_i > c$)
- Sample them during MCMC

Common Steps

Assume:

- Both observed and exact censored values come from the same distribution
- True value: $y_i|\beta_0, \beta_1, \sigma \sim \mathcal{N}(\mu_i, \sigma^2), \mu_i = \beta_0 + \beta_1 x_i$

For each data: Censored indicator: $\delta_i = 0$ if censored, 1 if observed

Solutions for Censored Data - Marginalization(1)

Core Idea: You never explicitly model the censored data y_i. Instead, you account for them by using the probability that the true value exceeds the censoring threshold $(y_i > c)$.

Likelihood for each data point:

- If observed $(\delta_i = 1)$: $p(y_i \mid \beta_0, \beta_1, \sigma)$
- If right-censored $(\delta_i = 0)$: $p(y_i > c \mid \beta_0, \beta_1, \sigma)$

Full Likelihood:

$$\mathcal{L}(\beta_0, \beta_1, \sigma) = \prod_{i:\delta_i=1} p(y_i \mid \beta_0, \beta_1, \sigma) \times \prod_{i:\delta_i=0} p(y_i > c \mid \beta_0, \beta_1, \sigma)$$

Solutions for Censored Data - Marginalization(2)

Likelihood for censored data ($\delta_i = 0$):

- If right-censored: $p(y_i > c \mid \beta_0, \beta_1, \sigma) = 1 - F(c \mid \beta_0, \beta_1, \sigma)$
- If left-censored: $p(y_i < c \mid \beta_0, \beta_1, \sigma) = F(c \mid \beta_0, \beta_1, \sigma)$
- If interval-censored: $p(a < y_i < b \mid \beta_0, \beta_1, \sigma) = F(b \mid \beta_0, \beta_1, \sigma) - F(a \mid \beta_0, \beta_1, \sigma)$

where $F(c \mid \beta_0, \beta_1, \sigma)$ as the CDF of the likelihood function

Pros:

- More efficient (no latent variables, fewer parameters), faster inference
- Better MCMC convergence/mixing

Cons:

- Requires simple closed-form CDF
- Harder for complex or custom models

Solutions for Censored Data - Data Augmentation

Core Idea: You explicitly introduce the missing values y_i and sample them.

Treat censored data as latent variables y_i^{mis} with constraint $y_i > c$

- $y_i^{mis} \sim \mathcal{N}(\mu_i, \sigma^2)$ and $y_i^{mis} > c$

Full Likelihood:

$$\mathcal{L}(\theta) = \prod_{i:\delta_i=1} p(y_i \mid \beta_0, \beta_1, \sigma) \times \prod_{i:\delta_i=0} p(y_i^{mis} \mid \beta_0, \beta_1, \sigma)$$

Solutions for Censored Data - Data Augmentation

Pros:

- Very flexible, works for complex models
- No need for CDF
- Provides imputed values for censored data

Cons:

- Slower (more parameters to sample)
- Can have poor mixing with many censored points

Applied Example: Lung Cancer Survival

Data: `survival::lung` in R. NCCTG study, $n = 228$ patients with advanced lung cancer.

Variables: `time` (days), `status` (1 = censored, 2 = died; 28% censored), age, `sex`, ECOG performance score.

Analysis model (Weibull):

$$\log T_i = \alpha + \beta_{\text{age}}\text{age}_i + \beta_{\text{sex}}\text{sex}_i + \sigma\epsilon_i, \quad \epsilon_i \sim \text{Gumbel}.$$

Questions:

- Does sex associate with longer survival, adjusting for age?
- What is the posterior median survival time, by sex?

Priors and Prior Predictive

Priors on the log-time scale:

$$\begin{aligned} \alpha &\sim \mathcal{N}(\log 365, 1), && \text{baseline median near 1 year} \\ \beta_{\text{age}}, \beta_{\text{sex}} &\sim \mathcal{N}(0, 0.5^2), && \text{weak shrinkage} \\ \text{shape} &\sim \text{Gamma}(2, 2), && \text{weakly informative.} \end{aligned}$$

Prior predictive: simulate T_i from (α, β, σ) drawn from priors, at observed covariates.

Targets:

- Predictive median survival between 2 months and 4 years.
- Tails exclude absurd values (millennia or seconds).
- Predictive 95% interval covers the observed range of times.

Marginalization: R Code

Marginalization via brms:

```
library(brms); library(survival);

# status: 1 = censored, 2 = observed event
lung$censored <- ifelse(lung$status == 1, "right", "none")

fit_marg <- brm(
  time | cens(censored) ~ age + sex,
  family = weibull(),
  data   = lung,
  prior  = c(prior(normal(5.9, 1), class = "Intercept"),
             prior(normal(0, 0.5), class = "b"),
             prior(gamma(2, 2),    class = "shape")),
  chains = 4, iter = 2000, seed = 5440)
```

Data Augmentation: Stan Code

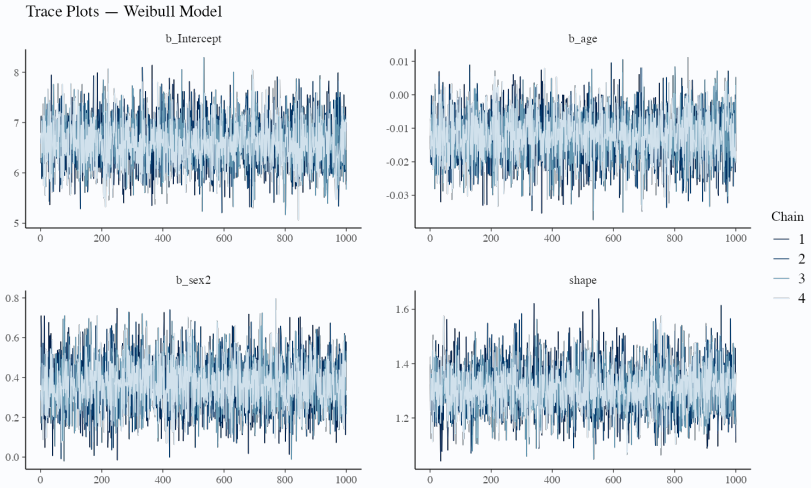
Data Augmentation via Stan:

```
parameters {
  real alpha;  vector[2] beta;  real<lower=0> shape;
  vector<lower=t_cens_lower>[N_cens] t_cens; // <-- latent event times
}
model {
  alpha ~ normal(5.9, 1); beta ~ normal(0, 0.5); shape ~ gamma(2, 2);

  // Observed: full Weibull density
  t_obs ~ weibull(shape, exp(alpha + X_obs * beta));

  // Censored: latent times sampled, constrained above c_i
  for (i in 1:N_cens)
    t_cens[i] ~ weibull(shape, exp(alpha + X_cens[i] * beta));
}
```

`vector<lower=t_cens_lower>` constrains each latent $t_{\text{cens}[i]}$ to exceed c_i. Slower than marginalization, but produces posterior draws of the imputed event times.

Data Marginalization: Fit and Diagnostics

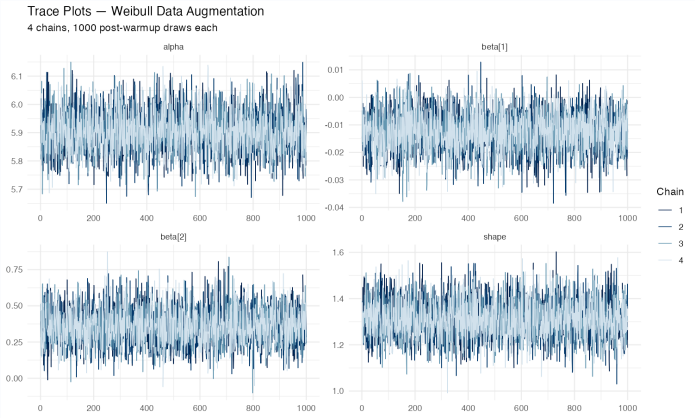
4 chains, 2000 iterations, warmup 1000. $\hat{R} < 1.01$ (max 1.002), Min ESS $= 0.483$, zero divergences.



Data Augmentation — Fit and Diagnostics

63 latent event times sampled as parameters alongside α, β, shape.

4 chains, 2000 iterations, warmup 1000. $\hat{R} \leq 1.0$, 0 divergences.



Trace plots — good mixing across all 4 chains

Posterior summaries:

M = Marginalization DA = Data Augmentation

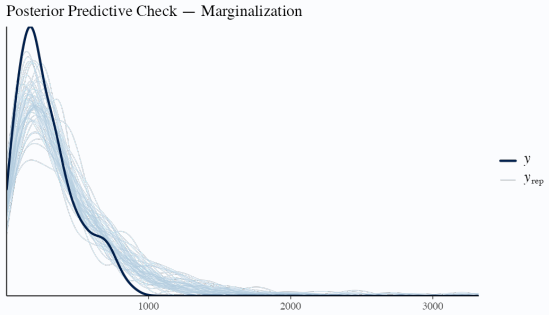
Parameter	M Mean	M 95% CrI	DA Mean	DA 95% CrI
β_{sex} (female vs. male, log-time)	0.37	(0.14, 0.62)	0.363	(0.13, 0.62)
β_{age} (per year)	-0.01	$(-0.03, 0.00)$	-0.012	$(-0.026, 0.001)$
Shape	1.31	(1.15, 1.47)	1.312	(1.15, 1.48)
Median survival, male (days)	255.7	(222.1, 297.2)	277.0	(239.0, 322.0)
Median survival, female (days)	367.8	(305.2, 453.5)	396.8	(327.6, 490.5)

Female sex associated with longer survival — β_{sex} CrI excludes zero. Consistent results between two approaches.

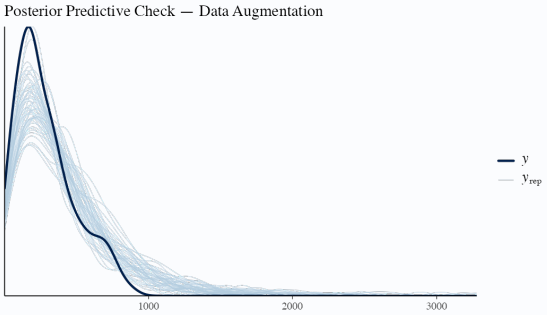
Posterior Predictive Check

Obstacle: y^{rep} cannot be compared to y on censored rows — observed value is only a lower bound.

Procedure: simulate event times, apply same censoring rule. Fraction censored: observed 27.6%, posterior 95% interval (50.4%, 64.0%).



Marginalization



Data Augmentation

Replicates match observed survival curve for first 1000 days. Weibull underpredicts the right tail in both cases.

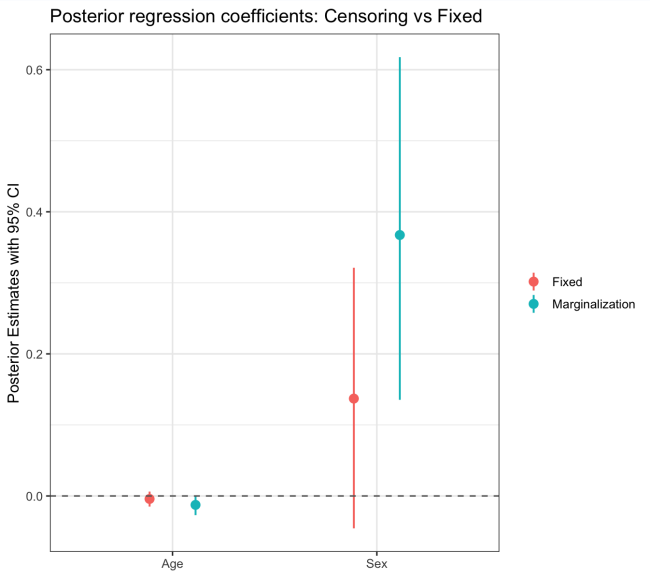
Sensitivity to the Parametric Family

The parametric family for the baseline distribution affects inference. **Refit under log-normal:**

Quantity	M Weibull	M Log-normal	DA Weibull	DA Log-normal
Median survival, male (days)	255.7	242.2	277.0	242.6
Median survival, female (days)	367.8	389.9	396.8	389.6
β_{sex} (log-time)	0.37 (0.14, 0.62)	0.48 (0.19, 0.78)	0.363 (0.13, 0.62)	0.472 (0.18, 0.76)

The positive female survival advantage holds under both families. Point estimates shift modestly; the key conclusion is robust.

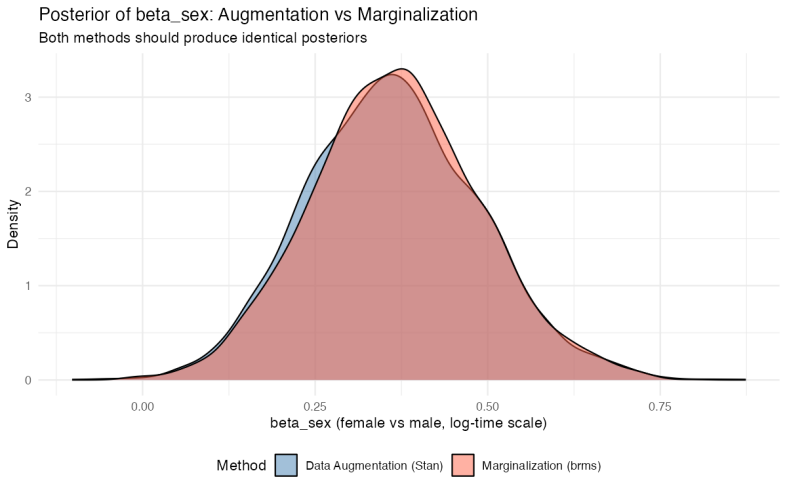
Importance of Dealing with Censored Data



- **Two models compared:**
  - Marginalization (correct censoring)
  - Fixed (treats censored as events)
- **Age effect:**
  - Close to zero in both models
  - No strong evidence of an age effect
  - Consistent across methods
- **Sex effect :**
  - Marginalization shows a stronger effect
  - 95% CI touches 0 for fixed model
  - Fixed model underestimates the effect size

Data Augmentation vs Marginalization

Both methods should produce identical posteriors for α, β, shape.



Marginalization
Mean: 0.367
CrI: (0.14, 0.62)

Augmentation
Mean: 0.363
CrI: (0.13, 0.62)

Only augmentation yields posterior draws of the 63 imputed event times.

Summary

Missing & Censored Data Commonalities

Both settings share the same three steps:

$$\textbf{augment} + \textbf{constrain} + \textbf{propagate}$$

- *Augment:* treat the unseen y_{mis} (or censored y_{cens}) as parameters.
- *Constrain:* by the joint model (missing data), additionally by $y_i > c_i$ (censored data).
- *Propagate:* the posterior over (θ, y_{mis}) induces the correct marginal posterior for θ.

Bayesian treat unknown data and unknown parameters the same way.

The Untestable Assumption

- Both problems rest on an assumption about *why* something is unseen.
  - Missing data: MAR. Missingness depends only on what is observed.
  - Censored data: non-informative censoring. Censoring time independent of event time given covariates.
- These assumptions concern structure outside the observed data. Then,PPCs and goodness-of-fit tests live on the observed part, so no posterior predictive check validates them.
- Solution:
  - Identify mechanisms in the data-collection process that could violate the assumption.
  - Run a sensitivity analysis across those mechanisms.
  - Report where the conclusion is robust and where it is not.

Takeaways

Missing Data Takeaways

- **Diagnose the Mechanism:** Identify the underlying nature of the missingness: *MCAR, MAR, or MNAR.*
- Predictors missing, mixed types $\Rightarrow$ `mice` + `brm_multiple`.
- Outcome missing, or joint posterior over (θ, y_{mis}) needed $\Rightarrow$ Stan data augmentation.
- MNAR concern $\Rightarrow$ selection or pattern-mixture model with sensitivity analysis.

Censor Data Takeaways

- Treating censored values as given both biases estimates and uses the wrong likelihood.
- If we have a closed-form CDF $\Rightarrow$ marginalization (`lccdf`, `brms cens()`).
- If we do not have a closed-form CDF, or posterior samples of latent times needed $\Rightarrow$ augmentation.
- Informative censoring $\Rightarrow$ joint model of event and censoring processes.

Bibliography

- Gelman, A. (2013) Bayesian data analysis (Chapters 8 & 18)
- Qing Zho. Lecture Notes UCLA- Bayesian Inference with Missing Data
- Datasets
  - NHANES
  - PLCO Lung

Appendix

Appendix- Ignorability Under MAR

Claim: Under MAR and $\theta \perp \phi$:

$$p(y_{\text{obs}}, I \mid \theta, \phi) \;=\; p(I \mid y_{\text{obs}}, \phi) \cdot p(y_{\text{obs}} \mid \theta).$$

Consequence for θ:

$$p(\theta \mid y_{\text{obs}}, I) \;\propto\; p(\theta)\, p(y_{\text{obs}} \mid \theta) \;=\; p(\theta) \int p(y_{\text{obs}}, y_{\text{mis}} \mid \theta)\, dy_{\text{mis}}.$$

The missingness mechanism ϕ drops out. We can *ignore* $p(I \mid y, \phi)$ and marginalize y_{mis} out.

Imputation is integral by Monte Carlo. Under MNAR, $p(I \mid y, \phi)$ depends on y_{mis}.

Appendix - Rubin's Rules

Given K imputed datasets, let $\hat{\theta}_k$ and $\hat{W}_k$ be the estimate and variance from dataset k.

- **Combined estimate:** $\bar{\theta}_K = \frac{1}{K}\sum_{k=1}^{K}\hat{\theta}_k$
- **Within-imputation variance:** $\bar{W}_K = \frac{1}{K}\sum_{k=1}^{K}\hat{W}_k$ — average uncertainty within each completed dataset
- **Between-imputation variance:** $B_K = \frac{1}{K-1}\sum_{k=1}^{K}(\hat{\theta}_k - \bar{\theta}_K)^2$ — variability across imputations, reflecting missingness uncertainty
- **Total variance:** $T_K = \bar{W}_K + \frac{K+1}{K}B_K$
- Interval estimates use a t-distribution based on T_K